College Football Analysis

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Introduction

- Goal: Determine whether offensive or defensive performance has a stronger impact on winning games

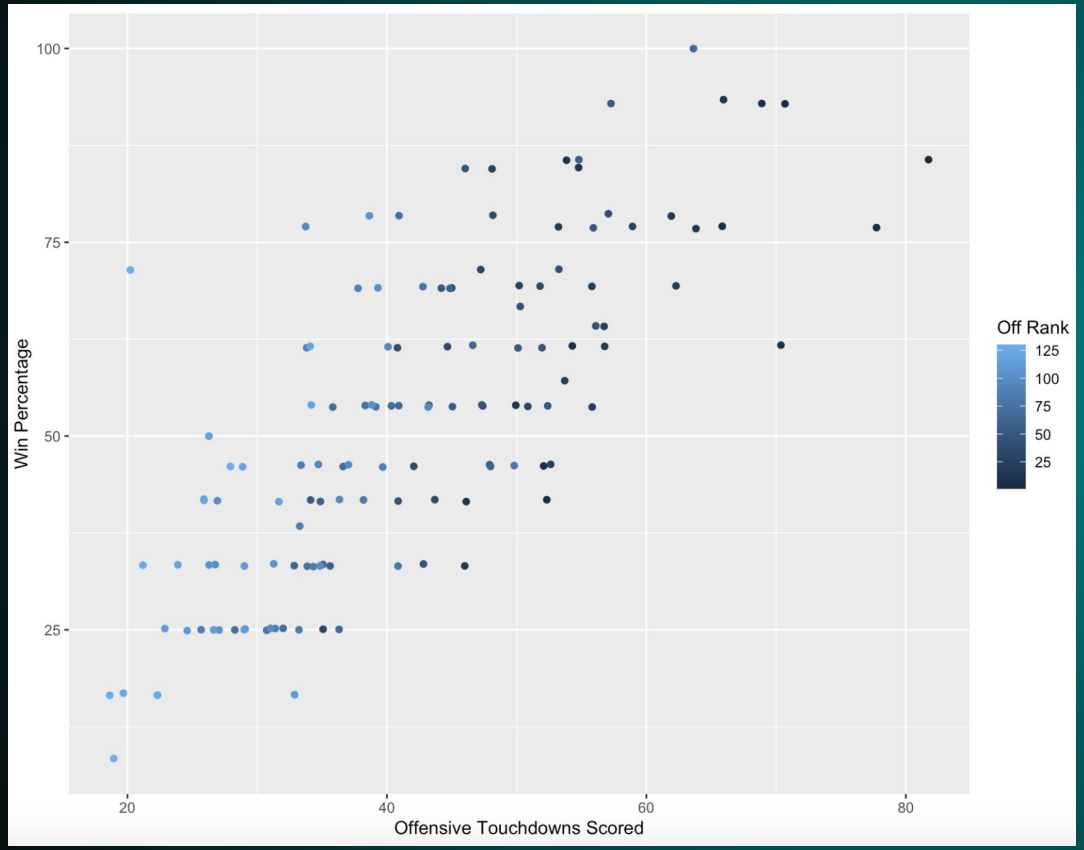
- Used the Kaggle data set for the 2023 College Football Season

Kaggle, https://www.kaggle.com/datasets/jeffgallini/college-football-team-stats-2019

Methodology

- Chose the 2023 college football season
- Cleaned the dataset
  - Seperated the Win-Loss variable into two columns, win and loss.
  - Converted non-numeric variables to numeric
  - Created percentage variable
- Created and used plots and linear regression to compare offense and defense
  - Touchdowns
  - Yardage
  - Turnovers

Offensive Touchdowns Scored



- Moderate positive linear relationship between offensive TDs scored and win percentage
- More TDs scored leads to a higher offensive rank

Offensive Touchdowns Allowed

- Moderate negative linear relationship between offensive TDs allowed and win percentage
- More TDs allowed leads to a lower defensive rank

Linear Models

Offensive TDs Scored

- For every offensive TD scored, win percentage increases by 1.22 points.
- ~58.2% of the variation in win percentage is explained by the number of touchdowns scored.
- The p-value is very small and shows statistical significance.
- Shows offensive touchdowns are predictive of wins.

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	0.870253	3.97703279	0.2188197	8.271311e-01
`Off TDs`	1.222022	0.09042206	13.5146412	1.324609e-26

2 rows

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
0.582331	0.5791427	13.54066	182.6455	1.324609e-26	1	-534.269	1074.538	1083.209

1 row | 1-9 of 12 columns

Offensive TDs Allowed

- For every offensive TD allowed, win percentage drops by 1.18 points.
- The model only explains ~28% of the variation.
- This is a weaker model than offensive TDs scored due to a lower level of explanation of variation and a higher p-value.
- Is still statistically significant, but not as much as TDs scored.

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	97.748736	6.5971586	14.816793	8.776731e-30
`Off TDs Allowed`	-1.178696	0.1660608	-7.097975	7.177777e-11

2 rows

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
0.2777644	0.2722511	17.80587	50.38124	7.177777e-11	1	-570.6885	1147.377	1156.048

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Offensive Touchdowns Allowed

- Teams that gain more yards tend to have higher win percentages (+0.203).
- Teams that allow more yards tend to have lower win percentages (-0.242).
- Both predictors are statistically significant since the p-values well below 0.05.
- ~62.6% of the variability in win percentage is explained by the model.

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
(Intercept)	64.3328259	11.09632273	5.797671	4.830972e-08
`Off Yards per Game`	0.2033465	0.01876546	10.836211	6.840158e-20
`Yards Per Game Allowed`	-0.2422484	0.02321455	-10.435198	6.822519e-19

3 rows

r.squared <dbl>	adj.r.squared <dbl>	sigma <dbl>	statistic <dbl>	p.value <dbl>	df <dbl>	logLik <dbl>	AIC <dbl>	BIC <dbl>
0.6262027	0.620452	12.85896	108.8911	1.664746e-28	2	-526.8891	1061.778	1073.34

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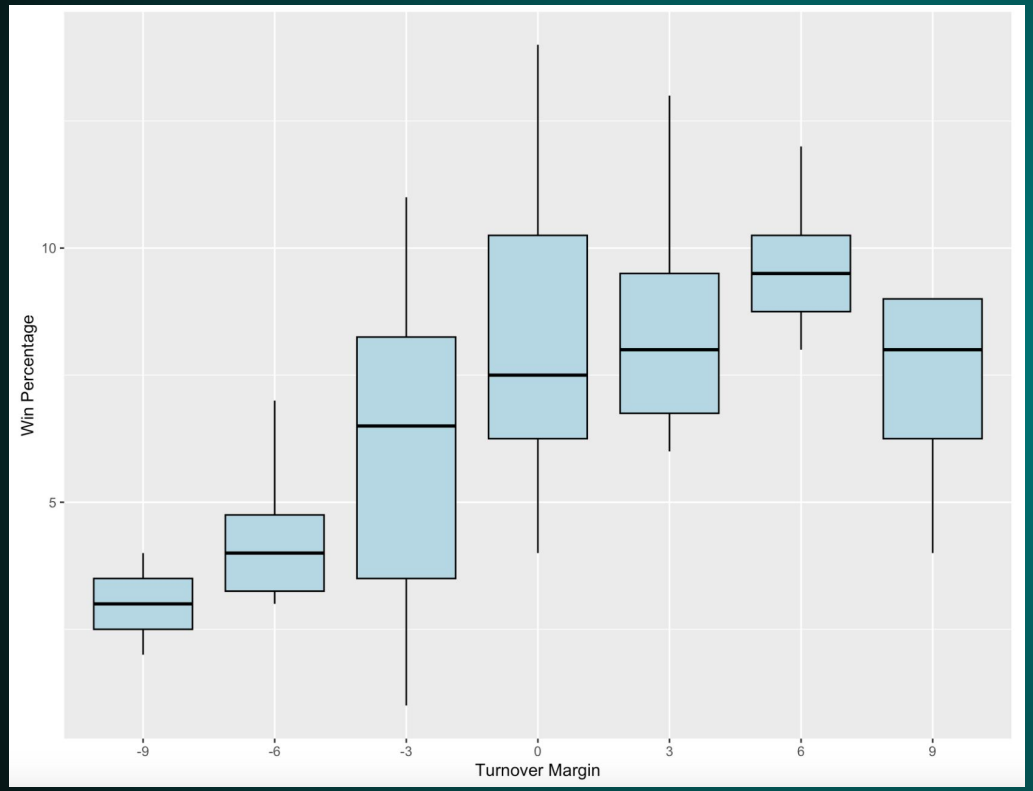
Turnover Margin



- Explains how well a team protects the ball on offense and creates turnovers on defense.
- A better turnover margin usually means the offense is turning the ball over less.
- Teams with better turnover margins tend to win more.

Turnover Margin

- For every increase in turnover margin, win percentage increases by 1.87 points.
- The p-value is < 0.05, making the relationship between turnover margin and win percentage statistically significant.
- ~34% of the variation in win percentage is explained by turnover margin.

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	51.46196	1.4791920	34.790592	4.908522e-68
`Turnover Margin`	1.87373	0.2282443	8.209321	1.836959e-13

2 rows

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
0.3396943	0.3346538	17.02536	67.39295	1.836959e-13	1	-564.7269	1135.454	1144.125

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Results and Conclusion

- Offensive TDs scored showed the strongest correlation with an R^2 of 0.582.
- Offensive yards per game was also highly predictive with a positive coefficient
- and high statistical significance.
- TDs allowed and yards allowed had negative correlations with win percentage.
- Turnover margin had a moderate relationship with win percentage.
- Variation kept decreasing as defensive variables were added to the linear models.

Offense is more reliable for winning games.